Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

(1) Let m be the Lebesgue measure on $[0,1]$ and $I_A(x)$ denote the indicator function of A.
(a) Consider for $n \geq 1$, $f_n(x) = nI_{[0,\frac{1}{n})}(x)$ and $f(x) = 0$ for $x \in [0,1]$. Show:
(i) $f_n \to f$ a.e(m).
(ii) f_n is **not** uniformly integrable (UI).
(iii) Check $f_n \not\to f$ in L^1.
(b) In general, under what conditions on the functions $\{f_n\}$ and f, would "$f_n \to f$ a.e (m)" imply "$f_n \to f$ in L^p $(0 < p < \infty)$"?
(c) Show that $f_n \to f$ in L^p implies that $f_n \to f$ in L^q, for any $0 < q < p < \infty$.
((3×6)+6+8=32 points)

(2) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.
(a) If $M(t) = E(e^{tX})$, $t \in I\!R$ is the moment generating function of X, then show

$$P(X > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$

(b) If $X \geq 0$ is random variable with $Var(X) < \infty$. Prove:

$$P(X > 0) \geq \frac{[E(X)]^2}{E(X^2)}.$$

[Hint: Use Cauchy-Schwarz Inequality] (8+10 =18 points)

(3) (a) **State** precisely the *Dominated convergence Theorem (DCT)* (Not the EDCT!).
(b) **Prove** that the conclusion of DCT remains true even if "$f_n \to f$ a.e(μ)" is replaced by "$f_n \to f$ in measure" in the statement. $(6 + 9 = 15$ points)

(4) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ-finite measures on it. Define the set function: $\mu_3 \equiv \mu_1 + \mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = \mu_1(A) + \mu_2(A)$). Then, prove the following:
$\mu_3(A) = \int_A d\mu_3 = \int_A \mu_1(dx) + \int_A \mu_2(dx)$
(a) μ_3 is a measure.
(b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.
(c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).
(d) If $\mu \ll \nu$ and $\mu \perp \nu$ holds, show that μ is the zero measure.
(e) If $\mu_1 = Lognormal(5,6), \mu_2 = Geometric(0.3)$ and $\nu = Poisson(8)$ on $(\Omega, \mathcal{F}) = (I\!R, \mathcal{B}(I\!R))$, then find the Lebesgue decomposition of μ_3 w.r.t. ν, and find the Radon Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

$\mu_{2s} = \mu_2(A \cap \{\mathbb{N}\}^c)$ (6+8+8+8+10=40 points)

$\mu_{2a} = \mu_2(A \cap \mathbb{N})$ $\mu_a = \mu_{2a}$

$\mu_{2a}(A) = \sum I_A(k)\, 0.3\, (0.7)^k$

$\mu_s = \mu_1 + \mu_{2s}$

$\mu_s(\mathbb{N}) = 0$ $\nu(\mathbb{N}^c) = 0$

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^{1} p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1,\ n \in \mathbb{N}$	$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b); $-\infty < a < b < \infty$	$m(A \cap [a, b])/(b-a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx,\ a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha,\beta)} \int_A x^{\alpha-1}(1-x)^{\beta-1} I_{(0,1)}(x) m(dx)$, where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b),\ a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R},\ \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R},\ \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2/2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R},\ \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2/2\sigma^2}}{x} I_{(0,\infty)}(x) m(dx)$

1. (a) (i) For $x = 1$, $f_n(1) = f(1) = 0$. So $f_n(1) \to f(1)$ as $n \to \infty$.
For $x = 0$, $f(0) = 0$ and $f_n(0) = n$, $\forall\ n$. So $f_n(0) \nrightarrow f(0)$ as $n \to \infty$.
For $x \in (0,1)$, then $\exists\ n_0$ s.t $\frac{1}{n_0} < x$ and $\forall\ n \geq n_0$, $f_{n_0}(x) = 0 = f(x)$. So $f_n(x) \to f(x), x \in (0,1)$ as $n \to \infty$.
So $\{x : f_n(x) \nrightarrow f(x)\} = \{0\}$ and $m(\{0\}) = 0$. Hence $f_n \to f$ a.e (μ).

(ii) Since $\int |f_n| I_{\{|f_n|>T\}}\ dm = nm\big([0, \frac{1}{n})\big) = 1$, $\forall\ n > T$, then $\sup_{n\geq 1} \int |f_n| I_{\{|f_n|>T\}}\ dm \geq 1$. So $\lim_{T\to\infty} \sup_{n\geq 1} \int |f_n| I_{\{|f_n|>T\}}\ dm \neq 0$. Hence f_n is not UI.

(iii) $\int |f_n - f|\ dm = \int n I_{[0,\frac{1}{n})}\ dm = n \cdot \frac{1}{n} = 1 \nrightarrow 0$. So $f_n \nrightarrow f$ in L^1.

(b) If $\{f_n^p\}$ is UI, i.e. $\lim_{T\to\infty} \sup_{n\geq 1} \int |f_n| I_{\{|f_n|>T\}}\ dm = 0$, then $f_n \to f$ a.e (m) would imply $f_n \to f$ in L^p (HW problem: 2.36(b)) since $m(\Omega) = m([0,1]) = 1 < \infty$.There are other correct answers too, for example, along the lines of DCT.

(c) Note that in this case $r = p/q > 1$ and hence $\phi(x) = x^r$, $x \in [0,1]$ is a convex function. Using Jensen's, we have $\left(\int |f_n - f|^q d\mu\right) \leq \left(\int |f_n - f|^{q \cdot \frac{p}{q}}\right)^{\frac{q}{p}} \left(\int 1\ d\mu\right)$

$$\left(\int |f_n - f|^q d\mu\right)^{\frac{p}{q}} \leq \left(\int |f_n - f|^{q\frac{p}{q}} d\mu\right) = \left(\int |f_n - f|^p d\mu\right).$$

or in other words, $\|f_n - f\|_q \leq \|f_n - f\|_p \to 0$, as $n \to \infty$.

2. (a) Using Markov's inequality, for each $t > 0$,

$$P(X > a) = P(e^{tX} > e^{ta}) \leq e^{-ta} M(t),$$

$P(e^{tX} \geq e^{ta}) \leq \frac{E(e^{tX})}{e^{ta}}$

and hence the result follows.

(b) Apply Cauchy-Schwarz to $XI_{\{X>0\}}$ to obtain

$$E[XI_{\{X>0\}}] \leq [E(X^2)]^{1/2} [P(X > 0)]^{1/2}$$

using the fact that $I^2_{\{X>0\}} = I_{\{X>0\}}$ and $E[I_{\{X>0\}}] = P(X > 0)$. Also, since X is nonnegative, $E(X) = E(XI_{(X=0)}) + E(XI_{(X>0)}) = E(XI_{(X>0)})$. Hence, squaring both sides above, we get $[E(X)]^2 \leq E(X^2)P(X > 0)$ and the result follows.

3. (a) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g$, $a.e\ (\mu), \forall\ n \geq 1$, $\int g\ d\mu < \infty$ and $f_n \to f\ a.e\ (\mu)$, then $f \in L^1(\Omega, \mathcal{F}, \mu)$, $\lim_{n\to\infty} \int f_n\ d\mu = \int f\ d\mu$ and $\lim_{n\to\infty} \int |f_n - f|\ d\mu = 0$.

(b) We need to prove the following modified DCT:
Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g, a.e\ (\mu), \forall\ n \geq 1$, $\int g\ d\mu < \infty$ and $f_n \to f$ in measure, then $f \in L^1(\Omega, \mathcal{F}, \mu)$, $\lim_{n\to\infty} \int f_n\ d\mu = \int f\ d\mu$ and $\lim_{n\to\infty} \int |f_n - f|\ d\mu = 0$.

Proof: First we state two facts in real analysis:
Fact 1: A sequence $\{x_n\}$ converges to x iff for any subsequence $\{x_{n_k}\}$ of $\{x_n\}$, there is a further subsequence $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ such that $x_{n_{k_l}} \to x$.
Fact 2 (Theorem 2.5.2): $f_n \to f$ in measure $\Rightarrow$ there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \to f$ $a.e$ (μ).
Then we prove $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu$.
Denote $a_n = \int f_n \, d\mu, a = \int f \, d\mu$ and $b_n = \int |f_n - f| \, d\mu$. To show $a_n \to a$ and $b_n \to 0$, we need to use Fact 1. Choose any subsequence $\{a_{n_k}\}$ of $\{a_n\}$, and for this subsequence, choose the corresponding subsequence functions $\{f_{n_k}\}$ of $\{f_n\}$. Since $f_n \to f$ in measure, then $\lim_{n\to\infty} \mu(|f_n - f| > \varepsilon) = 0$, $\forall \varepsilon > 0$, thus $\lim_{k\to\infty} \mu(|f_{n_k} - f| > \varepsilon) = 0$, $\forall \varepsilon > 0$, i.e. $f_{n_k} \to f$ in measure. By Fact 2, we can get a sub-subsequence $f_{n_{k_l}} \to f$ $a.e$ (μ). For this sub-subsequence $\{f_{n_{k_l}}\}$, conditions of DCT are true. Hence from DCT, we have $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{l\to\infty} \int f_{n_{k_l}} \, d\mu = \int f \, d\mu$, i.e. $a_{n_{k_l}} \to a$. By Fact 1, we get $a_n \to a$, i.e. $\lim_{n\to\infty} \int f_n \, d\mu = \int f \, d\mu$. Similarly, we have $\lim_{n\to\infty} \int |f_n - f| \, d\mu = 0$.

4. (a) $\mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0$;
$\mu_3(A) = \mu_1(A) + \mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall\ A \in \mathcal{F}$;
If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = \mu_1(\cup_i A_i) + \mu_2(\cup_i A_i) = \sum_i \mu_1(A_i) + \sum_i \mu_2(A_i) = \sum_i (\mu_1(A_i) + \mu_2(A_i)) = \sum_i \mu_3(A_i)$.
So μ_3 is a measure.

(b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = \mu_1(A) + \mu_2(A) = 0$. So $\mu_3 \ll \nu$.

(c) Claim: $\frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$ a.e (ν). *Proof.* Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists\ f_1, f_2 \geq 0$ s.t $\mu_1(A) = \int_A f_1 \, d\nu, \mu_2(A) = \int_A f_2 \, d\nu$. By definition, $\mu_3(A) = \mu_1(A) + \mu_2(A) = \int_A f_1 \, d\nu + \int_A f_2 \, d\nu = \int_A (f_1 + f_2) \, d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\frac{d\mu_3}{d\nu} = f_1 + f_2 = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$ a.e (ν).

(d) $\mu \perp \nu \Rightarrow \exists\ B \in \mathcal{F}$ s.t $\nu(B) = \mu(B^c) = 0$. Since $\mu \ll \nu$ and $\nu(B) = 0$, then $\mu(B) = 0$. So $\mu(B) = \mu(B^c) = 0 \Rightarrow \mu(\Omega) = 0$. Thus $\forall\ A \in \mathcal{F}, 0 \leq \mu(A) \leq \mu(\Omega) = 0 \Rightarrow \mu(A) \equiv 0$, $\forall\ A \in \mathcal{F}$, i.e. $\mu \equiv 0$.

(e) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $\mu_1(B) = 0 = \nu(B^c)$, i.e. $\mu_1 \perp \nu$. Choose $\mu_s = \mu_1, \mu_a = \mu_2$ and $\mu_3 = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\frac{d\mu_a}{d\nu}$.
Let $\mu_c(A) = |A \cap \mathbb{N}_0|$, $\forall\ A$ which defines a counting measure on $\mathbb{N}_0$. From table 4.6.1, $\nu(A) = 0 \Rightarrow A \cap \mathbb{N}_0 = \emptyset \Rightarrow \mu_a(A) = 0$, i.e. $\mu_a \ll \nu$. So μ_a is the absolute continuous part of μ_3 w.r.t. ν. Then $\mu_2(A) = \sum_{x\in\mathbb{N}_0} I_A(x)(0.7)^x 0.3 = \int p_G(x) d\mu_c(x)$ where $p_G(x) = I_{\mathbb{N}_0}(x)(0.7)^x 0.3$. Similarly $\nu(A) = \int p_P(x) d\mu_c(x)$ where $p_P(x) = e^{-8} 8^x (x!)^{-1} I_{\mathbb{N}_0}(x)$.

Since $p_P(x) > 0$ a.e (μ_c), then

$$\begin{aligned}
\frac{d\mu_a}{d\nu}(x) &= \frac{d\mu_2}{d\mu_c}\left(\frac{d\nu}{d\mu_c}\right)^{-1} \\
&= \left(I_{\mathbb{N}_0}(x)(0.7)^x 0.3\right) \cdot \left(I_{\mathbb{N}_0}(x)e^{-8}8^x(x!)^{-1}\right)^{-1} \\
&= \frac{(0.7)^x 0.3\ e^8 x!}{8^x} I_{\mathbb{N}_0}(x),\ a.e.(\nu).
\end{aligned}$$

(iii) For each $n \geq 1$, let μ^n be a probability measure on $(I\!R^n, \mathcal{B}(I\!R^n))$. Under what conditions can we say that there is a probability measure P on $(I\!R^\infty, \mathcal{B}(I\!R^\infty))$ such that $P(A \times I\!R^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(I\!R^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = I\!R^\infty, \mathcal{F} = \mathcal{B}(I\!R^\infty), P)$ such that the joint distribution of $(X_1, \ldots, X_n)$ is given by μ^n, for all $n \geq 1$.

(iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Normal(0,1)$ random variables. (4×7= **28** points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^{1} p^i (1-p)^{1-i} I_A(i)$
Binomial (n,p), $0 < p < 1$, $n \in \mathbb{N}$	$\sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A),\ A \in \mathcal{B}(\mathbb{R})$
Uniform (a,b), $-\infty < a < b < \infty$	$m(A \cap [a,b])/(b-a)$
Exponential (β), $\beta \in (0,\infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$
Gamma (α,β), $\alpha,\beta \in (0,\infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0,\infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx$, $a \in (0,\infty)$
Beta (α,β), $\alpha,\beta \in (0,\infty)$	$\frac{1}{B(\alpha,\beta)} \int_A x^{\alpha-1}(1-x)^{\beta-1} I_{(0,1)}(x) m(dx)$, where $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $a,b \in (0,\infty)$
Cauchy (γ,σ) $\gamma \in \mathbb{R}, \sigma \in (0,\infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2+(x-\gamma)^2} m(dx)$
Normal (γ,σ^2), $\gamma \in \mathbb{R}, \sigma \in (0,\infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2/\,2\sigma^2) m(dx)$ $\quad \nu = 0$
Lognormal (γ,σ^2), $\gamma \in \mathbb{R}, \sigma \in (0,\infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x-\gamma)^2/2\sigma^2}}{x} I_{(0,\infty)}(x) m(dx)$

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1. Let m be the Lebesgue measure on $[0,1]$ and $I_A(x)$ denote the indicator function of A. Consider the following sequence of random variables on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], P = m)$: For $n \geq 1$, $X_n(\omega) = nI_{(0,\frac{1}{n^2})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.
   (i) Show $X_n \to X$ almost surely (P).
   (ii) Does $X_n \to X$ in probability ?
   (iii) Show $X_n \to X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 2$, but this convergence does not hold for $2 \leq p < \infty$. (7+7+(7+7) = **28** points)

2. Let X, Y be two **independent** random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let m is the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2))$. Let f, g be two non-negative functions in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$.

   - (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove
   $$Var(X+Y) = Var(X) + Var(Y).$$
   - (b) Prove: $\sqrt{1 + (E(X))^2} \leq E\sqrt{1+X^2} \leq 1 + E(|X|)$.
   - (c) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove
   $$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f.g \quad a.e\ (m^2)$$
   $1 - \frac{2}{p} < 0$
   $1 < \frac{2}{p}$
   - (d) If f is continuous and bounded on $\mathbb{R}$, so is $f * g$.
   - (e) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e (m), f and g are as in (b) c.
   - (f) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) e say about the law of Z ?
   (8+8+8+8+8+(5+4)= **49** points)

3. (i) Show that a $Normal(0,1)$ random variable exists:
   In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to \mathbb{R}$, such that its c.d.f. is given by Φ where
   $$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du, \ x \in \mathbb{R}.$$
   (ii) Compute the following Lebesgue integral (Clearly stating any result used)
   $$EX = \int_\Omega X(\omega) dP(\omega).$$